nature portfolio

Peer Review File

Assimilative Causal Inference

Corresponding Author: Professor Nan Chen

This file contains all reviewer reports in order by version, followed by all author rebuttals in order by version.

Version 0:

Reviewer comments:

Reviewer #1

(Remarks to the Author)
Key results

The authors introduce Assimilative Causal Inference (ACI), a principled framework for uncovering causal relationships in dynamical systems using Bayesian data assimilation. A key strength of ACI is its ability to infer causal structure from a single realization, even when the underlying causal relationships evolve over time. Specifically, ACI adopts a Bayesian Inverse Problem perspective, asking whether knowledge of future observations helps reduce uncertainty about potential past causes. The authors also develop efficient algorithms to compute the causal influence range (CIR)—the time window over which a cause significantly impacts an effect. The method is demonstrated on simulated dynamical systems, highlighting its ability to recover transient and time-varying causal patterns.

Validity

The authors test the ACI framework on synthetic datasets generated by well-known dynamical systems, under the conditional gaussian nonlinear systems (CGNS) assumptions on the data generation process. While the analysis of the results is sound and highlights the key features of the method well, there is no explicit accuracy measure of the inferred causal influences or CIRs. Comparing ACI's outputs against a ground-truth dataset with explicitly known causal influences and CIRs would help validate the method's accuracy and address concerns about the reliability of its inferred causal relationships and CIR estimates.
The manuscript's claim of scalability to high-dimensional systems is not supported by theoretical analysis or empirical evidence. To substantiate this claim, the authors should either provide a discussion of the method's computational complexity as a function of the number of variables, or demonstrate its practical performance on higher-dimensional datasets. Currently, all experiments are limited to systems with at most six state variables, which does not convincingly demonstrate scalability.

Significance

This work has potentially high significance, as it addresses several major limitations of existing temporal causal discovery methods. In principle, ACI can accommodate instantaneous causal effects, latent confounders, and time-varying causal structures, all from a single realization. However, these capabilities remain largely theoretical, as they are not fully explored or validated through empirical analysis. Additional experiments are needed to validate the robustness of the proposed method against these factors. Additionally, the manuscript lacks comparisons with established baseline methods, such as causation entropy, transfer entropy, or convergent cross mapping (CCM), which limits the ability to assess ACI's relative strengths and practical utility.

Data and methodology

All experiments are conducted on low-dimensional, synthetic datasets generated by the authors, including benchmark systems such as the nonlinear Dyad model and a simplified ENSO model. While these examples effectively demonstrate the method's core ideas, they offer limited insight into ACI's performance on real-world data, which typically involve higher dimensionality, observation noise, and potential violations of the CGNS assumptions. As no real-world datasets are analyzed, and important factors like sensitivity to noise, model mismatch, or partial observability are not systematically examined, it is difficult to assess the method's robustness and generalizability. Including an experiment on a real-world dataset would substantially strengthen the methodological contribution.

Analytical approach

The analytical framework proposed in the paper is creative and well thought out. The derivation of estimators under the Conditional Gaussian Nonlinear System (CGNS) assumption is detailed and appears sound, though I did not verify all low-level derivations. The use of infinite uncertainty for the non-target variables (i.e.., x_B) is a subtle but intuitive modeling choice, and the justification provided aligns well with Bayesian data assimilation principles.
However, the framework's heavy reliance on the CGNS assumption needs to be examined in more detail. Since the tractability of ACI hinges on this assumption, it would be helpful to understand how critical it truly is. How does performance degrade when the system violates this condition? An empirical test of the method under such violations would strengthen the argument for its broader applicability.
Additionally, while ACI offers a fresh perspective by tracing causes backward from observed effects, it appears conceptually similar to transfer entropy and causation entropy, both of which also quantify the information flow between time-lagged variables. A discussion contrasting ACI with these methods—particularly in terms of expressivity, computational complexity, and assumptions—would help position the contribution more clearly within the existing literature. Without such a comparison, it remains unclear when and why ACI would be preferred over these well-established alternatives.

Clarity and context

While the overall presentation is clear, certain technical aspects would benefit from additional context and clarification. For example, the discussion in Section 4 regarding the use of infinite uncertainty for x_B is difficult to follow in isolation. Its rationale becomes clear only when cross-referenced with Supporting Information Sections 1.4.2 and 1.4.3.
Additionally, the computational procedure underlying ACI is fairly complex. Including a pseudocode algorithm block that outlines the major steps would improve accessibility for a broader audience and facilitate reproducibility.

Suggested improvements

Validation and Ground Truth Testing: Include experiments with synthetic datasets where the true causal influences and CIRs are explicitly known.
Scalability Assessment: Provide theoretical analysis of computational complexity as a function of system dimensionality, and demonstrate ACI's performance on higher-dimensional datasets.
Robustness Analysis: Conduct systematic experiments examining ACI's sensitivity to key factors including observation noise, violations of CGNS assumptions, and partial observability.
Baseline Comparisons: Include comparative analysis with established methods such as causation entropy, transfer entropy, and convergent cross mapping (CCM).
Real-world Application: Add at least one experiment on real-world data to demonstrate practical applicability beyond synthetic examples.
Enhanced Clarity: Include a pseudocode algorithm block outlining ACI's major computational steps to improve accessibility and reproducibility.

References

The manuscript appropriately references most relevant works in the field. To strengthen the literature review, consider adding a citation to Graber and Schwing (2020) [1], which specifically addresses dynamic causal links using Granger causality and provides methods for causal discovery from single realizations.
[1] Graber, Colin, and Alexander Schwing. "Dynamic neural relational inference for forecasting trajectories." Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops. 2020.

Reviewer #2

(Remarks to the Author)
I co-reviewed this manuscript with one of the reviewers who provided the listed reports. This is part of the Nature Communications initiative to facilitate training in peer review and to provide appropriate recognition for Early Career Researchers who co-review manuscripts.

Reviewer #3

(Remarks to the Author)
Dear editor and authors,

This is my review of the manuscript entitled "Assimilative Causal Inference".

The manuscript introduces, formally defines and thoroughly tests a new method for inferring causality: Assimilative Causal

Influence (ACI). ACI promises to detect causal influence under a variety of conditions, including high-dimensional systems, even in cases where the strength and direction of causality changes over time. These conditions are often requirements in fields such as climatology, ecology and neuroscience and presumably also highly relevant in other areas such as physiology and economics. In addition to ACI the paper introduces Causal Influence Range (CIR), which is a measure for the time-scale at which the the causal influence is effective. The authors circumvent an inherent ambiguity (subjectivity) in this measure by developing an objective version which is a normalized integral over all values of relative entropy.

ACI is based on inverse inference, i.e., knowing future values of the effect (at time later than t) reduces the uncertainty in inferring the value of the cause at time t. This is formalised using relative entropy (KL divergence) in a Bayesian data assimilation framework. The method is clearly explained in the main paper, with a more detailed explanation in the SI. Figure 2.1 is also very helpful for understanding ACI and objective CIR.

My overall assessment is that this is a very well-written paper and that the developed method is novel and will be highly interesting and relevant to many researchers. The authors present their method and results in a succinct and thorough way, and my recommendation is to publish the paper more or less as is, except for one suggestion and some minor issues (see below).

Major comment:
I really only have one major suggestion, which is that I think the authors should make the code used for analysis and figures available up-front, instead of requiring interested readers to contact the corresponding author. I think this will be in the interest of the authors as well, since it will make the method more accessible. A second reason is that I have had several unfortunate experiences of writing authors to request code and/or data from papers and either have not received an answer or been rejected on various grounds. While I have no reason to think that the authors of the current manuscript would not be open to sharing, sometime people change jobs, e-mail addresses or even lose or misplace code and data. I sincerely hope the authors will follow this suggestion. This is a beautiful and well organized paper, and I think having the code would increase its usefulness to interested readers a lot.

Minor comments:
– In the abstract the authors write that ACI "provides online tracking..." I was initially confused about the use of the word online and prefer the formulation they use on page 1, line 39: "real-time tracking of the causal relationship" which, if I understand correctly, means the same thing.
– On p. 5, l. 173 the wrong symbol is used. It should be ε, not ϵ for the threshold parameter.
– On p. 7, l. 274 the same mistake recurs.
– In section 2.3 the variable x is split into two subsets x_A and x_B, but this is not mentioned explicity (except in the SI). It would be helpful to state briefly in the main text as well, since x is used later on in the section to refer to the union of x_A and x_B (e.g. on p. 6, l. 234-235).
– In section 3.1, Eq. (3.1a) and (3.1b): W_x and W_y (and their derivatives) are not defined or explained. Again the SI will help the reader, but a short comment here would help.
– Figure 3.1: If space permits, it would be interesting to see a third panel with a plot of objective CIR as a function of t. I find it a bit difficult to decode the whisker plot in panel a.

Regards,
Dan Mønster

Reviewer #4

(Remarks to the Author)
Review: Assimilative Causal Inference, by Andreou, Chen and Bollt

The authors introduce a framework for Assimilative Causal Inference (ACI), which harnesses Bayesian data assimilation to trace the candidate causes of observations. It is demonstrated to be effective for high-dimensional dynamical systems and short datasets, providing real-time tracking of causal roles and establishing a mathematically rigorous criterion for the causal influence range.

The authors describe the general significance of ACI across various scientific disciplines and some of the limitations of existing methods in capturing instantaneous, time-evolving causal relationships in complex systems. The approach is both powerful in scope of application and rigorous in a self-consistent sense and thus should be of interest to a broad scope of researchers interested in complex dynamical systems. In my opinion the paper should be published following a more sober assessment of its limitations and comparison/contrast to other approaches, such as those described in the second paragraph of the introduction. For example, I find it intellectually and emotionally distasteful for an author to claim their approach is “paradigm-shifting”, as is done in the second sentence of the abstract, upon its introduction? Paradigm shifts require a broad adoption and consensus in the community and years of testing and refinement. This concept was introduced by Thomas Kuhn in his “The Structure of Scientific Revolutions”, Univ Chicago Press, 1962. Paradigm shifts in science are not defined by fiat by an author and they take time to evolve. How do the authors know that next week, month or year, etc. another method will not appear that is far superior and supersede all others? Indeed, let the community and time decide what is paradigm shifting.

The paper highlights the potential of ACI to open new avenues for studying complex systems with transient causal structures and suggests further areas for research and application of ACI. It is one important method of identifying and tracing back causes from observed effects through Bayesian data assimilation and can identify interactions without direct candidate cause observations. It can provide real-time tracking and mathematically validated causal influence range and was shown to be effective in systems showcasing intermittency and extreme events.

As is noted, there are both general limitations and application-specific limitations, that make the paradigm-shifting claims specious. These include:

(1) The fact that the ACI framework may be affected by the correlation between variables during Bayesian data assimilation, which can potentially introduce a spurious causal relationship, even if conditioned over intermediary variables. This can create erroneous interpretations of data.

(2) The real-time tracking capability of the ACI framework necessitates continuous observation of a dataset. This requirement can be challenging in systems where data is not continuously available or is incomplete, which is often the case.

(3) Although ACI claims to scale efficiently to high dimensions, the complexity of the framework still poses significant computational challenges, particularly in very large-scale systems.

(4) The framework assumes the underlying models are correctly specified without observational noise, a condition that is rarely true in real-world applications where measurements are often noisy in many distinct, but difficult to quantify, ways.

(5) While the authors noted the applicability in short datasets, the accuracy and reliability of causal inferences drawn from very limited data remain a concern.

(6) The application-specific limitations include (a) extreme event analysis, where accurate parameter selection and model calibration are essential; (b) historical data limitations, such as shifts not represented in a dataset and (c) interference handling, which can compromise the reliability of the inferred causal relationships.

While these limitations are noted, and notable, the reader will strongly benefit from knowing how these few, of potentially many more, are treated in other approaches in the field, many of which are mentioned in the introduction and are known to the authors in more detail.

Version 1:

Reviewer comments:

Reviewer #1

(Remarks to the Author)
I would like to thank the authors for their detailed response and changes. I believe that the new version of the paper is substantially better. However, some of my concerns remain:

Regarding claims of scalability: I appreciate the authors' clarification that ACI is designed to leverage existing Bayesian data assimilation machinery and hence their method can be applied in high-dimensional settings. However, such clarification is not well supported. The rebuttal does not provide a concrete complexity analysis of the composite ACI method in terms of state dimension, nor does it include new experiments on higher-dimensional systems. The current experiments still only contain low-dimensional examples (up to six variables), which does not convincingly substantiate the strong claim that the method "scales efficiently to high-dimensional problems."
I would therefore encourage the authors, at minimum, to soften the wording of the scalability claim throughout the paper. Perhaps the authors can emphasize that ACI can in principle be implemented in high-dimensional settings by reusing efficient DA algorithms, while leaving a full empirical demonstration and more detailed complexity analysis to future work.

"By interpolating causality from effects to causes instead of extrapolating the latter forward-in-time ACI offers an inherently more robust framework compared to classical predictive causality."

While I understand the key difference in approach, I do not see how it is apparent that "interpolating causality from effects to causes" results in an "inherently more robust framework". Without sufficient theoretical or empirical justification, especially reading up to the point of the paper where the claim is made, it is hard to validate it.

Regarding real-world experiments: I appreciate the authors' response highlighting the difficulty of working with issues like addressing model error and sampling biases. I also believe that the new experiments with the real-world ENSO data have made the paper stronger.

However, I would still encourage the authors to add a discussion to compare the experimental results of their method to the baselines on the real-world ENSO dataset to highlight scenarios where their method would perform more favorably compared to the baselines. While the theoretical contributions of the paper are indeed quite significant, practically demonstrating the ability of ACI to discover instantaneous edges and the resulting improvements for causal discovery performance, would further strengthen the contribution of the paper.

Reviewer #2

(Remarks to the Author)
I co-reviewed this manuscript with one of the reviewers who provided the listed reports. This is part of the Nature Communications initiative to facilitate training in peer review and to provide appropriate recognition for Early Career Researchers who co-review manuscripts.

Reviewer #3

(Remarks to the Author)
Dear editor and authors,

This is my response to the revised manuscript entitled "Assimilative Causal Inference."

I want to thank the authors for a very thorough, detailed and easy-to-read response letter and for providing diff-files making it clear what has been changed.

After reviewing the revised manuscript, I am satisfied that my initial comments have all been adequately addressed by the authors. Specifically, I am happy to see that the autjors have provided a repository with the code used to produce the results in the manuscript.

I recommend publication of this revised manuscript, and I look forward to more research on the method and to see applications by other researchers.

Regards,
Dan Mønster

Reviewer #4

(Remarks to the Author)
I thank the authors for their thorough response to my report and those of the other referees and the associated revisions of the text, and am pleased to recommend the paper for publication.

Response to Reviewers

Ref.: Response to Referees
Manuscript Number: NCOMMS-25-48940
Manuscript Title: Assimilative Causal Inference
Authors: Marios Andreou; Nan Chen; Erik Bollt
Corresponding Author: Nan Chen

Dear Reviewers:

We deeply appreciate the reviewers for taking the time to thoroughly review our work and provide such thoughtful and insightful feedback. The reviewers' detailed and discerning eye helped us clarify several valuable points and implement additions and revisions that will further aid readers in understanding and appreciating our framework. We sincerely thank the reviewers for their time, rigor, insights, and endorsement through the peer review process.

The revised manuscript and Supplementary Information (SI) are attached for your consideration. All changes are also clearly marked in the attached `diff` files, both for the main text and SI, which we upload alongside our revised manuscript and response letter. These files clearly indicate the changes made to our initial submission. Specifically, blue text indicates additions made in the revision, while ~~strikethrough~~ red text indicates removals.

In the following pages, we provide a detailed, point-by-point response to the reviewers' comments, outlining how each has been thoughtfully considered in the revised manuscript. At the end, we also list some additional minor changes that have been made to the initial manuscript beyond the reviewers' recommendations and suggestions. These include fixing minor typos, adjusting figures for clarity and readability, and making additional clarifications to the exposition. These revisions do not alter the theoretical results or the deductions made from the numerical case studies carried out in the initial submission. We hope that with these additional changes and corrections, our manuscript is much more readable and accessible.

What follows is a detailed account of how each of the reviewers' points has been addressed in the revised manuscript and Supplementary Information (SI).

Point-by-Point Response to Reviewers' Comments:

Reviewer #1

Remarks to the Authors:

Key results

The authors introduce Assimilative Causal Inference (ACI), a principled framework for uncovering causal relationships in dynamical systems using Bayesian data assimilation. A key strength of ACI is its ability to infer causal structure from a single realization, even when the underlying causal relationships evolve over time. Specifically, ACI adopts a Bayesian Inverse Problem perspective, asking whether knowledge of future observations helps reduce uncertainty about potential past causes. The authors also develop efficient algorithms to compute the causal influence range (CIR)—the time window over which a cause significantly impacts an effect. The method is demonstrated on simulated dynamical systems, highlighting its ability to recover transient and time-varying causal patterns.

► **Authors' Response:** We sincerely thank the reviewer for taking the time to provide a very thorough and constructive review of our paper. In what follows, we carefully outline how each concern raised has been addressed in the revised manuscript. We believe that the changes made in line with the reviewer's suggestions have substantially improved the clarity, rigor, and overall quality of the paper. For that, we again thank the reviewer.

Validity

The authors test the ACI framework on synthetic datasets generated by well-known dynamical systems, under the conditional Gaussian nonlinear systems (CGNS) assumptions on the data generation process.

► **Authors' Response:** Thank you for pointing this out. We would like to offer an important clarification: although the numerical illustrations of ACI in this work use dynamical systems with the CGNS structure, the ACI framework itself is developed for general nonlinear dynamical systems (see also Section 1.2.1 in the SI). The CGNS models were chosen as test cases because their closed-form analytic expressions in Bayesian data assimilation eliminate the sampling errors that typically arise from ensemble methods. This allows us to focus on studying the ACI framework itself without the confounding influence of sampling or other numerical errors. The general formulation of dynamical systems under the ACI framework is as follows:

$$\begin{aligned} \mathrm{d}\mathbf{x}(t) &= \mathbf{f}^{\mathbf{x}}(t,\mathbf{x},\mathbf{y})\mathrm{d}t + \boldsymbol{\Sigma}_1^{\mathbf{x}}(t,\mathbf{x})\mathrm{d}\mathbf{W}_1(t) + \boldsymbol{\Sigma}_2^{\mathbf{x}}(t,\mathbf{x})\mathrm{d}\mathbf{W}_2(t), \\ \mathrm{d}\mathbf{y}(t) &= \mathbf{f}^{\mathbf{y}}(t,\mathbf{x},\mathbf{y})\mathrm{d}t + \boldsymbol{\Sigma}_1^{\mathbf{y}}(t,\mathbf{x},\mathbf{y})\mathrm{d}\mathbf{W}_1(t) + \boldsymbol{\Sigma}_2^{\mathbf{y}}(t,\mathbf{x},\mathbf{y})\mathrm{d}\mathbf{W}_2(t). \end{aligned} \tag{1}$$

In (1), we do not assume that $\mathbf{f}^{\mathbf{x}}$ or $\mathbf{f}^{\mathbf{y}}$ are linear in the unobserved variable $\mathbf{y}$, or that the noise feedback matrices $\Sigma_1^{\mathbf{y}}$ and $\Sigma_2^{\mathbf{y}}$ do not depend on $\mathbf{y}$, which are the assumptions for a CGNS. The only assumptions that accompany (1) are the (I)–(V) regularity conditions noted in Section 1.2.1 of the SI. We understand that this modeling assumption is not explicitly communicated to the reader in the main text and appears only in the SI, which is a concern that Reviewer #3 has also raised. Therefore, we have added the following paragraph to the beginning of Section 2.1.1 in the main text to explicitly address that the ACI framework developed here applies to general stochastic systems, not limited to CGNSs:

"Consider a time interval $[0,T]$ and a specific time instant t, where $0 \le t \le T$. Let $\{\mathbf{x}(t)\}_{0\le t\le T}$ and $\{\mathbf{y}(t)\}_{0\le t\le T}$ be two multivariate stochastic processes defined over this interval. Denote by $\mathbf{x}(t)$ and $\mathbf{y}(t)$ the corresponding multivariate random variables at time t. Their evolution in time is defined by the following stochastic dynamical system:

$$\begin{aligned} \frac{\mathrm{d}\mathbf{x}}{\mathrm{d}t} &= \mathbf{f}^{\mathbf{x}}(t,\mathbf{x},\mathbf{y}) + \boldsymbol{\Sigma}_1^{\mathbf{x}}(t,\mathbf{x})\dot{\mathbf{W}}_1 + \boldsymbol{\Sigma}_2^{\mathbf{x}}(t,\mathbf{x})\dot{\mathbf{W}}_2, \\ \frac{\mathrm{d}\mathbf{y}}{\mathrm{d}t} &= \mathbf{f}^{\mathbf{y}}(t,\mathbf{x},\mathbf{y}) + \boldsymbol{\Sigma}_1^{\mathbf{y}}(t,\mathbf{x},\mathbf{y})\dot{\mathbf{W}}_1 + \boldsymbol{\Sigma}_2^{\mathbf{y}}(t,\mathbf{x},\mathbf{y})\dot{\mathbf{W}}_2, \end{aligned} \tag{2.1}$$

where $\dot{\mathbf{W}}_1$ and $\dot{\mathbf{W}}_2$ are independent mean-zero Gaussian random vectors with uncorrelated components that possess unit variance. Mathematical details regarding the structure of (2.1) are given in the Supplementary Information."

While the analysis of the results is sound and highlights the key features of the method well, there is no explicit accuracy measure of the inferred causal influences or CIRs. Comparing ACI's outputs against a ground-truth dataset with explicitly known causal influences and CIRs would help validate the method's accuracy and address concerns about the reliability of its inferred causal relationships and CIR estimates.

► **Authors' Response:** Thank you very much for raising these important concerns. With respect to explicit CIR accuracy benchmarks, to the best of our knowledge, ACI is the first general framework designed to quantify the temporal reach of instantaneous causal impact in the predictive (future) direction. Consequently, it differs fundamentally from traditional time-averaged causal inference methods, in which the availability of large datasets enables the definition of a statistical ground truth and the implementation of standard validation procedures. In contrast, the goal of the ACI framework is to detect instantaneous causal relationships. In this setting, the true signal consists of only a single data point at each time instant, which makes conventional statistical validation (such as direct comparison with ground-truth or theoretical CIR values) impractical, even for relatively simple models. For this reason, such traditional validation is not included in the present work.

Nevertheless, we introduced the dyad model with extreme events as our first test example. Since the mechanism triggering extreme events is apparent from the dynamical structure, we used this underlying dynamical property as a qualitative truth to validate our ACI method. Similarly, in the SI, the well-understood feedback loop mechanism in the predator–prey model is used to validate the inference results from the ACI method. We also wish to point out that several of the tests in this work are conducted under perfect-model scenarios, meaning there is no model error when validating the ACI method. In addition, closed-form analytic formulae from the CGNSs studied also eliminate sampling errors present in ensemble methods. These setups provide an idealized environment for validating the method's accuracy. We have added the following at the beginning of the results section to clarify that we used these well-understood models to generate synthetic data for validating ACI and to point out the difference between the standard validation approaches:

"The dynamical properties of these systems and the mechanisms that trigger crucial observed phenomena are well understood. This knowledge is exploited as guidance to validate the results when applying ACI. In addition, a perfect-model setup is used in the case studies carried out, meaning the system generates the synthetic observational data and the forecast model in ACI at the same. This facilitates the recovery of causal relationships in the absence of model error. Because the underlying system is chaotic and subject to stochastic forcing, a perfect model means that the model used in ACI shares the same governing equations and long-term statistical behavior as the system that generates the true signal. Additional results, including further numerical experiments, additional case studies, and one crucial application using sparse real-world observations, are provided in the Supplementary Information."

Overall, the crucial role of the CIR played in understanding intermittent extreme events (Sections 3.1 and SI.4.1), random cause–effect role switches (Section SI.4.2), and conditional causal relationships in realistic models of complex Earth processes (Sections 3.2 and SI.4.3) collectively provide supportive evidence of ACI's ability to capture temporal causal influences, as the framework's outputs align well with established physical understandings. This is further illustrated in the revised Supplementary Figure 4.2, which clarifies the underlying mechanisms and the interplay between the subjective and objective CIRs (see point also (4) at the end). In addition, the newly added case study using real ENSO data (Section 4.3.3 in the revised SI) offers further validation, thanks to the reviewer's helpful suggestion.

The manuscript's claim of scalability to high-dimensional systems is not supported by theoretical analysis or empirical evidence. To substantiate this claim, the authors should either provide a discussion of the method's computational complexity as a function of the number of variables, or demonstrate its practical performance on higher-dimensional datasets. Currently, all experiments are limited to systems with at most six state variables, which does not convincingly demonstrate scalability.

► **Authors' Response:** We thank the reviewer for raising this question. ACI exploits Bayesian data assimilation, which allows it to leverage the computationally efficient algorithms and techniques developed in this mature field. These tools enable the method to efficiently perform the required computations and store auxiliary components (e.g., filter covariances, gain operators). In other words, scalability is supported by the broad suite of computational strategies already established in Bayesian data assimilation.

Part of this is demonstrated through the use of the adaptive-lag online smoother for CGNSs, which is employed in this work to efficiently estimate the objective CIRs at each time step (see Sections 2.2–2.3 of the SI). Yet, since the technique details are in the SI, we have summarized this point in the revised main text. When ensemble-based methods are used, the goal is typically to obtain an approximate representation of the solution rather than the exact distribution. In practice, the ensemble size (often on the order of $O(100)$) is chosen largely independently of the system's dimension. The associated approximation error is, of course, an important topic, but it is distinct from the present study and warrants a separate thorough investigation.

Following the reviewer's suggestion, to provide discussions on the method's computational complexity, we have added the following sentence in the introduction:

"Third, ACI scales efficiently to high-dimensional problems by leveraging computational methods developed in Bayesian data assimilation. This is fundamentally different from information-transfer methods limited to low dimensions."

We have also added the following remarks addressing this point in the conclusion section:

*"Second, by building on the computationally efficient methods developed in Bayesian data assimilation, it is essential to apply ACI to higher-dimensional systems. Ensemble data assimilation is a well-established technique that routinely uses localization, covariance inflation, and other adjustments to overcome the curse of dimensionality [**54, 55, 56**]. As a result, the ensemble size (typically on the order of $O(100)$) is often chosen independently of the system's dimension, which contributes to the scalability of Bayesian data assimilation and, consequently, of ACI. However, the sampling errors inherent in ensemble methods may introduce additional biases. Thus, it is necessary to examine whether such errors affect ACI. In addition, various coarse-grained approaches, such as approximating the full distribution with a Gaussian, are widely used in practice. Although these approximations introduce errors in computing the PDF, they are robust and highly accurate when computing diagnostic indicators, such as the relative entropy, that are used for model identification and sensitivity analysis. Thus, they may substantially accelerate ACI in practice."*

Significance

This work has potentially high significance, as it addresses several major limitations of existing temporal causal discovery methods. In principle, ACI can accommodate instantaneous causal effects, latent confounders, and time-varying causal structures, all from a single realization. However, these capabilities remain largely theoretical, as they are not fully explored or validated through empirical analysis. Additional experiments are needed to validate the robustness of the proposed method against these factors. Additionally, the manuscript lacks comparisons with established baseline methods, such as causation entropy, transfer entropy, or convergent cross mapping (CCM), which limits the ability to assess ACI's relative strengths and practical utility.

► **Authors' Response:** First, we would like to thank the reviewer for their constructive input and for recognizing ACI's potential to address key limitations of current methods in temporal causal discovery, particularly its ability to accommodate instantaneous causal effects, latent confounders, and time-varying causal structures from just a single partial realization.

We additionally thank the reviewer for raising these helpful questions. In this work, we used the dyad model with extreme events, the predator–prey model, and a nonlinear model for the El Niño–Southern Oscillation (ENSO) in a perfect-model setting to study instantaneous causal effects, latent confounders, and time-varying causal structures, all from a single realization. The underlying dynamical properties of these models are well understood, which allows us to use them to validate the results produced by ACI. The method showed strong consistency across multiple challenging topics, including extreme events, intermittency, regime switching, and multiscale interactions. Because ACI relies on only one time series, meaning we have only one data point at each time instant, knowledge of the underlying dynamics is crucial for these validation tests. Moreover, in all examples we used a perfect-model setup and a nonlinear data assimilation algorithm with closed-form analytic expressions. As a result, there is no model error and no sampling error (as would arise with ensemble methods), and therefore an empirical analysis of such error sources is not needed here. We wish to emphasize that this paper focuses on the methodology and the rigorous theoretical justification of the ACI framework. The specifically designed nonlinear and non-Gaussian test cases further illustrate rich features of complex natural phenomena. Practical issues, many of which depend on the application, will be addressed in several follow-up papers, as outlined above and in the discussion section.

Following the reviewer's suggestion, we have also added an additional example in the SI that uses real ENSO data to validate the method (Section 4.3.3 in the revised SI). This case is no longer a perfect-model test. Thus, it provides a useful demonstration of the robustness of the method and offers initial evidence that ACI remains valid under realistic observational conditions (noisy and sparse/limited data) and tolerates a reasonable amount of model error.

Since ACI focuses on instantaneous causal detection and the calculation of instantaneous causal inference ranges, it addresses different aspects of causality compared to causation entropy, transfer entropy, or CCM. These are excellent methods with many valuable applications, but they primarily characterize overall or time-aggregated causal relationships. Consequently, they are not suitable benchmarks for validating ACI's instantaneous causal detection or instantaneous causal ranges, and we therefore do not perform direct comparisons. We appreciate the reviewer raising this point and have clarified this distinction in the discussion sections, which provides a qualitative comparison of the intended uses of these methods:

"Finally, ACI requires only a single time series but assumes access to a dynamical model, which enables it to identify instantaneous causal relationships. In contrast, most time-series-based methods operate under model-free assumptions and therefore focus on aggregated or time-averaged causal links rather than on causality at specific moments. Because ACI incorporates model dynamics, it does not require long time series to detect causes and effects. Instead, it can infer instantaneous causal links from a single data point combined with model information. In contrast, most purely data-driven methods rely on long time series and can only reveal causal relationships in a time-averaged sense."

We note that a potential, but limited, form of comparison in this direction would be to time-average the ACI or CIR outputs to obtain empirical indicators. However, such averaging removes the unique instantaneous information that ACI is designed to capture. Moreover, the metrics used by the different methods are fundamentally different, which may further complicate any meaningful comparison.

Data and methodology

All experiments are conducted on low-dimensional, synthetic datasets generated by the authors, including benchmark systems such as the nonlinear Dyad model and a simplified ENSO model. While these examples effectively demonstrate the method's core ideas, they offer limited insight into ACI's performance on real-world data, which typically involve higher dimensionality, observation noise, and potential violations of the CGNS assumptions. As no real-world datasets are analyzed, and important factors like sensitivity to noise, model mismatch, or partial observability are not systematically examined, it is difficult to assess the method's robustness and generalizability. Including an experiment on a real-world dataset would substantially strengthen the methodological contribution.

► **Authors' Response:** Thank you for bringing this up. Our goal with this paper is to introduce the ACI framework and demonstrate its effectiveness, performance, and robustness across a range of demanding scenarios, including intermittent extreme events, causal-role reversals, and complex real-world geophysical phenomena. Since the methodology already involves many crucial points and the results include interesting findings, we wish to focus on clarifying these points. The framework will eventually be applied to more realistic systems, along the lines of what the reviewer points out which we sincerely appreciate. However, as we responded previously, extending the framework to the entire set of practical cases requires substantial additional work and careful considerations, including addressing model error, sampling biases, and other issues the reviewers pointed out. This may deserve another two or three entire papers to study. We have added many discussions of these points, as the reviewer suggested.

We also acknowledge the reviewer's suggestion and have therefore included an experiment on real-world data, which will strengthen ACI's methodological contribution by accounting for observations that are impartial, noisy, incomplete/irregular, observed at discrete intervals, and not necessarily stemming from the underlying dynamical model. Following the reviewer's suggestion, we have added a new subsection to the SI of this work, Section 4.3.3 in the revision: "ACI on Real-World ENSO Data", where we apply the conditional ACI framework on the conceptual ENSO model in (4.3) with observations defined by daily (for τ) and monthly (for all other variables) data for the state variables. The observations come from the following reanalysis datasets (instead of using model-generated values):

1. NCEP Global Ocean Data Assimilation System (GODAS)
2. NCEP-NCAR Reanalysis 1 Project

We have also updated our "Data Availability Statement" in the main text accordingly. The details can be found in this new subsection in the revised SI. We have also summarized the key findings at the end of the corresponding results section in the main text:

"In addition, in the Supplementary Information, a case study using real-world ENSO observations is examined, which does not assume a perfect model setup. The conclusions derived about the causal relationships

using real-world observational data remain qualitatively similar to those with model-generated data. The results provide preliminary evidence that the ACI framework remains reasonably robust under realistic conditions, including observational noise, moderate model error, and numerical discretization effects."

Analytical approach

The analytical framework proposed in the paper is creative and well thought out. The derivation of estimators under the Conditional Gaussian Nonlinear System (CGNS) assumption is detailed and appears sound, though I did not verify all low-level derivations. The use of infinite uncertainty for the non-target variables (i.e., $\mathbf{x}_{\mathrm{B}}$) is a subtle but intuitive modeling choice, and the justification provided aligns well with Bayesian data assimilation principles. However, the framework's heavy reliance on the CGNS assumption needs to be examined in more detail. Since the tractability of ACI hinges on this assumption, it would be helpful to understand how critical it truly is. How does performance degrade when the system violates this condition? An empirical test of the method under such violations would strengthen the argument for its broader applicability.

► **Authors' Response:** We would like to thank the reviewer for the positive and encouraging comments regarding our proposed methodology for assessing unconditional and conditional instantaneous causal relationships. We also thank the reviewer for pointing out this issue, as readers may otherwise misunderstand the applicability of the ACI method. Although the test models used in this work belong to the CGNS class, the ACI framework itself is designed for general nonlinear dynamical systems. As explained previously, we use CGNSs in the test cases to create a perfect-model scenarios with closed-form analytical expressions, thereby avoiding potential numerical errors and allowing us to focus entirely on the results produced by ACI without concerns about numerical biases. This choice helps clarify and validate the method. To make this point explicit, we have added equations and explanations at the beginning of the methods section (Section 2 in the main text) and clearly emphasized that the ACI framework is developed for general nonlinear systems (see (1)).

Additionally, while ACI offers a fresh perspective by tracing causes backward from observed effects, it appears conceptually similar to transfer entropy and causation entropy, both of which also quantify the information flow between time-lagged variables. A discussion contrasting ACI with these methods—particularly in terms of expressivity, computational complexity, and assumptions—would help position the contribution more clearly within the existing literature. Without such a comparison, it remains unclear when and why ACI would be preferred over these well-established alternatives.

► **Authors' Response:** We thank the reviewer for describing our proposed inverse-problem-based approach as a fresh perspective on causality. Because ACI reformulates classical predictive causality under Granger and Wiener as a Bayesian inverse problem through data assimilation, it differs substantially from other established causal inference approaches. This includes transfer entropy, causation entropy, and the additional methods explicitly mentioned in Section 1 of the main text, such as information flow, convergent cross mapping, and Koopman causality. We have highlighted these distinctions in the methods sections:

"Under (2.5), ACI provides a unique formulation for identifying causal relationships. In traditional approaches, causal links are inferred by running (2.1) forward to quantify the informational response on the target variables' entropy when conditioned on the history of the candidate causes. ACI instead solves an inverse problem, thus tracing causes back from the data of the observed subsequent effects. By interpolating causality from effects to causes instead of extrapolating the latter forward-in-time, ACI offers an inherently more robust framework compared to classical predictive causality."

We have also added a paragraph in the discussion section that further contrasts ACI with these other methods explicitly in terms of expressivity, computational complexity, and assumptions.

"It is worth highlighting how ACI differs from other state-of-the-art methods in terms of expressivity, computational complexity, and underlying assumptions. Different from most methods that identify cause-and-effect relationships by running the system forward and quantifying informational responses in the target variables' entropy conditioned on past candidate causes, which is a predictive setting typical of information-flow-based

*methods, ACI instead formulates causality as an inverse problem in uncertainty quantification. Within the Bayesian data assimilation framework, candidate causal variables are inferred backward from observations of their potential effects. Thus, while many existing methods extrapolate causes forward in time, ACI traces causes back from observed effects, effectively interpolating rather than extrapolating causality. This distinction makes ACI more expressive in capturing instantaneous causal structures and, in practice, often more robust. From a computational standpoint, ACI can leverage well-developed Bayesian data assimilation tools that scale effectively to high-dimensional systems [**53, 36**]. In contrast, information transfer and other model-based methods often require computing the exact density functions and their derivatives, which is computationally demanding and subject to the curse of dimensionality. Finally, ACI requires only a single time series but assumes access to a dynamical model, which enables it to identify instantaneous causal relationships. In contrast, most time-series-based methods operate under model-free assumptions and therefore focus on aggregated or time-averaged causal links rather than on causality at specific moments. Because ACI incorporates model dynamics, it does not require long time series to detect causes and effects. Instead, it can infer instantaneous causal links from a single data point combined with model information. In contrast, most purely data-driven methods rely on long time series and can only reveal causal relationships in a time-averaged sense."*

Clarity and context

While the overall presentation is clear, certain technical aspects would benefit from additional context and clarification. For example, the discussion in Section 4 regarding the use of infinite uncertainty for $\mathbf{x}_\mathrm{B}$ is difficult to follow in isolation. Its rationale becomes clear only when cross-referenced with Supporting Information Sections 1.4.2 and 1.4.3.

► **Authors' Response:** Thank you for bringing up this concern. Indeed, (we assume the reviewer here is referring to Section 2.3 of the manuscript) the exposition and discussion in Section 2.3 of the main text regarding the use of infinite uncertainty for the non-target variables $\mathbf{x}_\mathrm{B}$ when developing the theory of conditional ACI can be difficult to follow in isolation. Following the reviewer's recommendation, for self-containedness and better readability, we have revised part of Section 2.3 in the main text to the following:

"In the more general context of a complex system, state variables can be grouped into three disjoint subsets: $\mathbf{x}_\mathrm{A}$, $\mathbf{x}_\mathrm{B}$, and $\mathbf{y}$, with $\mathbf{x} = (\mathbf{x}_\mathrm{A}, \mathbf{x}_\mathrm{B})^\mathtt{T}$ under the previous notation. The goal is to determine whether $\mathbf{y}$ is a cause of $\mathbf{x}_\mathrm{A}$ in the presence of the remaining variables $\mathbf{x}_\mathrm{B}$, which are called the non-target variables, including the cases of confounders or mediators. Since $\mathbf{x}_\mathrm{B}$ interacts with both $\mathbf{x}_\mathrm{A}$ and $\mathbf{y}$, appropriately accounting for the role of $\mathbf{x}_\mathrm{B}$ while inferring the causal link from $\mathbf{y}$ to $\mathbf{x}_\mathrm{A}$ becomes essential. Hereafter, let us assume that a realization for each of $\mathbf{x}_\mathrm{A}$ and $\mathbf{x}_\mathrm{B}$, namely $\mathbf{x}_\mathrm{A}(s \le T)$ and $\mathbf{x}_\mathrm{B}(s \le T)$, is available. Therefore, the candidate cause $\mathbf{y}$ is the only variable that does not require to have observations in this framework. Following the argument in Section 2.1, the causal relationship from $\mathbf{y}(t)$ to $(\mathbf{x}_\mathrm{A}, \mathbf{x}_\mathrm{B})$ is identified based on the uncertainty reduction in the smoother distribution p_t^s related to the filter one p_t^f, where

$$
\begin{aligned}
p_t^\mathrm{f} &= p(\mathbf{y}(t)|\mathbf{x}_\mathrm{A}(s \le t), \mathbf{x}_\mathrm{B}(s \le t)),\\
p_t^\mathrm{s} &= p(\mathbf{y}(t)|\mathbf{x}_\mathrm{A}(s \le T), \mathbf{x}_\mathrm{B}(s \le T)).
\end{aligned}
$$

The main focus here is to appropriately determine and resolve the presence of $\mathbf{x}_\mathrm{B}$ from p_t^f and p_t^s as to infer a conditional causal relationship from $\mathbf{y}(t)$ to $\mathbf{x}_\mathrm{A}$ beyond the contributions of $\mathbf{x}_\mathrm{B}$.

*Recall from Section 2.1 that ACI employs Bayesian data assimilation to estimate the state of $\mathbf{y}(t)$. ACI exploits observations of $\mathbf{x}$ to reduce the uncertainty in $\mathbf{y}(t)$, where the influence of $\mathbf{x}$ on the estimation of $\mathbf{y}(t)$ depends on its own uncertainty: smaller observational uncertainty leads to a stronger impact [**45**]. When extending this framework to $\mathbf{x} = (\mathbf{x}_\mathrm{A}, \mathbf{x}_\mathrm{B})^\mathtt{T}$ for conditional ACI, the observed time series of $\mathbf{x}_\mathrm{B}$ is still used in the forecast step to compute the uncertainties in $\mathbf{x}_\mathrm{A}$ and $\mathbf{y}$. However, during the analysis step, the uncertainty in $\mathbf{x}_\mathrm{B}$'s dynamics is deliberately excluded by assigning infinite uncertainty to its marginal likelihood before posterior computation. This can be rigorously formulated through a limit operation on the analysis component of the Bayesian data assimilation update equations, which formally leads to the following filter- and smoother-based distributions:*

$$
p_t^{\mathrm{f}|\mathbf{x}_\mathrm{B}} := \lim_{\mathrm{Var}(\mathbf{x}_\mathrm{B}(t)) \to +\infty} p_t^\mathrm{f}, \quad p_t^{\mathrm{s}|\mathbf{x}_\mathrm{B}} := \lim_{\mathrm{Var}(\mathbf{x}_\mathrm{B}(t)) \to +\infty} p_t^\mathrm{s}.
$$

Similar to (2.5), if the relative entropy of $p_t^{\mathrm{s}|\mathbf{x}_{\mathrm{B}}}$ from $p_t^{\mathrm{f}|\mathbf{x}_{\mathrm{B}}}$ is positive, $\mathcal{P}\big(p_t^{\mathrm{s}|\mathbf{x}_{\mathrm{B}}}, p_t^{\mathrm{f}|\mathbf{x}_{\mathrm{B}}}\big) > 0$, then $\mathbf{y}$ is identified as the cause of $\mathbf{x}_{\mathrm{A}}$ conditional to $\mathbf{x}_{\mathrm{B}}$ at time t. This means future data of $\mathbf{x}_{\mathrm{A}}$ incur additional information gain about $\mathbf{y}(t)$ related to its history, beyond the observational contributions of $\mathbf{x}_{\mathrm{B}}$. By computing this metric at each t, a time-dependent conditional causal relationship is established."

Additionally, the computational procedure underlying ACI is fairly complex. Including a pseudocode algorithm block that outlines the major steps would improve accessibility for a broader audience and facilitate reproducibility.

► **Authors' Response:** This is an excellent suggestion! Thank you for the advice. This suggestion, as well as the one of Reviewer #3 to make our codebase publicly available, will significantly improve the accessibility and reproducibility of this work. We have made all MATLAB m-file scripts used to reproduce the analyses and figures in the main text and SI available at the following GitHub repository: `https://github.com/marandmath/ACI_code`. To further facilitate the understanding by general audience, we also implement the reviewer's suggestion to provide pseudocode algorithm blocks for the following:

1. Base ACI framework skeleton: See Section 1.3 in the SI (Algorithm 1).
2. Conditional ACI method: See Section 1.4.2 in the SI (Algorithm 2).
3. Computation of the subjective and objective CIRs, including the latter's computationally efficient approximation, for CGNSs: See Section 2.3 in the SI (Algorithm 3).

Since space is limited in the main text, all pseudocode blocks are in the SI.

We have chosen to include pseudocode for the computation of the CIRs specifically for CGNSs so readers have a concrete implementation they can work with instead of the abstract theory from Section 1.5 of the SI. Nevertheless, we note here that the pseudocode provided in Algorithm 3 is easily applicable to the general system in (1) by appropriately computing $\delta(t_n; t_j) = \mathcal{P}_n^j$ for each $j \in [\![N]\!]$ and $n \in \{j, \ldots, N\}$.

Suggested improvements

- Validation and Ground Truth Testing: Include experiments with synthetic datasets where the true causal influences and CIRs are explicitly known.
- Scalability Assessment: Provide theoretical analysis of computational complexity as a function of system dimensionality, and demonstrate ACI's performance on higher-dimensional datasets.
- Robustness Analysis: Conduct systematic experiments examining ACI's sensitivity to key factors including observation noise, violations of CGNS assumptions, and partial observability.
- Baseline Comparisons: Include comparative analysis with established methods such as causation entropy, transfer entropy, and convergent cross mapping (CCM).
- Real-world Application: Add at least one experiment on real-world data to demonstrate practical applicability beyond synthetic examples.
- Enhanced Clarity: Include a pseudocode algorithm block outlining ACI's major computational steps to improve accessibility and reproducibility.

► **Authors' Response:** We again thank the reviewer for taking the time to provide to us such a thorough and constructive review full of important suggestions and feedback. Please see our previous responses to see how we addressed each of these suggested improvements. We believe that with the implementations and changes made here, we have significantly improved the quality of the manuscript.

References

The manuscript appropriately references most relevant works in the field. To strengthen the literature review, consider adding a citation to Graber and Schwing (2020) [1], which specifically addresses dynamic causal links using Granger causality and provides methods for causal discovery from single realizations.

[1] Graber, Colin, and Alexander Schwing. "Dynamic neural relational inference for forecasting trajectories". Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops. 2020.

► **Authors' Response:** Thank you for the positive comments and for the suggestion. We have included the suggested reference in the second paragraph of the introduction (Section 1 in the main text).

Reviewer #2

Remarks to the Authors:

I co-reviewed this manuscript with one of the reviewers who provided the listed reports. This is part of the Nature Communications initiative to facilitate training in peer review and to provide appropriate recognition for Early Career Researchers who co-review manuscripts.

► **Authors' Response:** We sincerely thank the reviewer and their co-reviewer for their efforts and constructive feedback provided under the Nature Communications co-review initiative for early career researchers. Their input and time is greatly appreciated.

Reviewer #3

Remarks to the Authors:

Dear editor and authors,

This is my review of the manuscript entitled "Assimilative Causal Inference".

The manuscript introduces, formally defines and thoroughly tests a new method for inferring causality: Assimilative Causal Influence (ACI). ACI promises to detect causal influence under a variety of conditions, including high-dimensional systems, even in cases where the strength and direction of causality changes over time. These conditions are often requirements in fields such as climatology, ecology and neuroscience and presumably also highly relevant in other areas such as physiology and economics. In addition to ACI the paper introduces Causal Influence Range (CIR), which is a measure for the time-scale at which the the causal influence is effective. The authors circumvent an inherent ambiguity (subjectivity) in this measure by developing an objective version which is a normalized integral over all values of relative entropy.

ACI is based on inverse inference, i.e., knowing future values of the effect (at time later than t) reduces the uncertainty in inferring the value of the cause at time t. This is formalised using relative entropy (KL divergence) in a Bayesian data assimilation framework. The method is clearly explained in the main paper, with a more detailed explanation in the SI. Figure 2.1 is also very helpful for understanding ACI and objective CIR.

My overall assessment is that this is a very well-written paper and that the developed method is novel and will be highly interesting and relevant to many researchers. The authors present their method and results in a succinct and thorough way, and my recommendation is to publish the paper more or less as is, except for one suggestion and some minor issues (see below).

► **Authors' Response:** We sincerely thank the reviewer for their very positive and encouraging assessment of our manuscript. We are grateful for their recognition of the novelty, clarity, and relevance of our proposed causal inference method, as well as of their thoughtful suggestion and proposed minor corrections. We have carefully addressed all the points raised, making corresponding revisions and additions to the manuscript. We believe these changes have further improved the clarity and overall quality of the paper and its accompanying material. Our detailed responses to each comment are provided below.

Major comment:

I really only have one major suggestion, which is that I think the authors should make the code used for analysis and figures available up-front, instead of requiring interested readers to contact the corresponding author. I think this will be in the interest of the authors as well, since it will make the method more accessible. A second reason is that I have had several unfortunate experiences of writing authors to request code and/or data from papers and either have not received an answer or been rejected on various grounds. While I have no reason to think that the authors of the current manuscript would not be open to sharing, sometime people change jobs, e-mail addresses or even lose or misplace code and data. I sincerely hope the authors will follow this suggestion. This is a beautiful and well organized paper, and I think having the code would increase its usefulness to interested readers a lot.

► **Authors' Response:** This is an excellent suggestion! We fully agree that open access to code enhances transparency, reproducibility, and accessibility. The examples brought up by the reviewer serve as great reasons to why that needs to be norm for methodology papers. In response to this suggestion, we have now made all code used for the analyses and the production of the figures for all of the case studies carried out in this work publicly available at the following GitHub repository:

`https://github.com/marandmath/ACI_code`

We have also updated our "Code Availability Statement" in the main text accordingly. We have chosen to share our code in MATLAB, as it strikes a good balance between accessibility and fast performance.

Further following the reviewer's suggestion for wider accessibility and increased usefulness to interested readers, in this repository we have made a conscious effort to meticulously document all of the code in each script file with explanatory comments. We have also added a thorough docstring at the beginning of each m-file, both for exposition and for further implementation instructions.

In addition to allowing users to reproduce the analyses and figures present in this work, we have also added code which generates results and experiments not present in it (due to length constraints); for example subjective and objective CIR comparison plots for all case studies, and plots comparing the objective CIR definition and its computationally efficient approximation over time. We also include curated daily and monthly data from the following reanalysis datasets: NCEP Global Ocean Data Assimilation System (GODAS) and NCEP-NCAR Reanalysis 1, specific to the ENSO conceptual model. These can be used by the user in lieu of observations generated from the model simulation as a way to conduct a more realistic ACI analysis of the ENSO diversity.

We believe that with this addition we have increased the usefulness of our work to a wide range of practitioners and theorists. We again thank the reviewer for this very useful suggestion, as well as for their very flattering comments regarding our manuscript.

Minor comments:

- In the abstract the authors write that ACI "provides online tracking..." I was initially confused about the use of the word online and prefer the formulation they use on page 1, line 39: "real-time tracking of the causal relationship" which, if I understand correctly, means the same thing.

  ► **Authors' Response:** Indeed, we have used "real-time" here to refer to "online", as in our use of "online" in Section 2.2 of the SI regarding the online smoother for CGNSs, to denote the real-time smoother algorithm in the scenario where observations are obtained sequentially over time. Overall, "online" is a more precise term. Therefore, we have revised the abstract and changed "real-time" with "online", per the reviewer's suggestion. We have done the same for another similar instance of the use of "real-time" from the introduction.

- On p. 5, l. 173 the wrong symbol is used. It should be ε, not ϵ for the threshold parameter.
- On p. 7, l. 274 the same mistake recurs.

► **Authors' Response:** Thank you so much for catching these! We have replaced both instances of ϵ (`\epsilon`) with ε (`\varepsilon`), including two more instances that happened in the SI.

- In section 2.3 the variable $\mathbf{x}$ is split into two subsets $\mathbf{x}_\mathrm{A}$ and $\mathbf{x}_\mathrm{B}$, but this is not mentioned explicitly (except in the SI). It would be helpful to state briefly in the main text as well, since $\mathbf{x}$ is used later on in the section to refer to the union of $\mathbf{x}_\mathrm{A}$ and $\mathbf{x}_\mathrm{B}$ (e.g. on p. 6, l. 234–235).

► **Authors' Response:** Thank you for bringing this to our attention. We have now clarified in the revision of Section 2.3 in the main text that $\mathbf{x} = (\mathbf{x}_\mathrm{A}, \mathbf{x}_\mathrm{B})^\intercal$. Here is the revised excerpt: "In the more general context of a complex system, state variables can be grouped into three disjoint subsets: $\mathbf{x}_\mathrm{A}$, $\mathbf{x}_\mathrm{B}$, and $\mathbf{y}$, with $\mathbf{x} = (\mathbf{x}_\mathrm{A}, \mathbf{x}_\mathrm{B})^\intercal$ under the previous notation.".

- In section 3.1, Eq. (3.1a) and (3.1b): $\mathbf{W}_\mathbf{x}$ and $\mathbf{W}_\mathbf{y}$ (and their derivatives) are not defined or explained. Again the SI will help the reader, but a short comment here would help.

► **Authors' Response:** Thank you for bringing this up. Indeed, we have not concretely defined Wiener processes in the main text for brevity. We have added the following exposition to the beginning of Section 2.1.1 in the main text which briefly defines and explains this concept to the reader independent of the SI.

"Consider a time interval $[0, T]$ and a specific time instant t, where $0 \leq t \leq T$. Let $\{\mathbf{x}(t)\}_{0\leq t\leq T}$ and $\{\mathbf{y}(t)\}_{0\leq t\leq T}$ be two multivariate stochastic processes defined over this interval. Denote by $\mathbf{x}(t)$ and $\mathbf{y}(t)$ the corresponding multivariate random variables at time t. Their evolution in time is defined by the following stochastic dynamical system:

$$\begin{aligned}\frac{\mathrm{d}\mathbf{x}}{\mathrm{d}t} &= \mathbf{f}^\mathbf{x}(t,\mathbf{x},\mathbf{y}) + \boldsymbol{\Sigma}_1^\mathbf{x}(t,\mathbf{x})\dot{\mathbf{W}}_1 + \boldsymbol{\Sigma}_2^\mathbf{x}(t,\mathbf{x})\dot{\mathbf{W}}_2,\\ \frac{\mathrm{d}\mathbf{y}}{\mathrm{d}t} &= \mathbf{f}^\mathbf{y}(t,\mathbf{x},\mathbf{y}) + \boldsymbol{\Sigma}_1^\mathbf{y}(t,\mathbf{x},\mathbf{y})\dot{\mathbf{W}}_1 + \boldsymbol{\Sigma}_2^\mathbf{y}(t,\mathbf{x},\mathbf{y})\dot{\mathbf{W}}_2,\end{aligned} \tag{2.1}$$

where $\dot{\mathbf{W}}_1$ and $\dot{\mathbf{W}}_2$ are independent mean-zero Gaussian random vectors with uncorrelated components that possess unit variance. Mathematical details regarding the structure of (2.1) are given in the Supplementary Information."

- Figure 3.1: If space permits, it would be interesting to see a third panel with a plot of objective CIR as a function of t. I find it a bit difficult to decode the whisker plot in panel (a).

► **Authors' Response:** As a way to succinctly convey the concept of the forward- or future-pointing objective CIR, while minimizing the length of the main text, we have decided to use that whisker plot convention in Figure 3.1 to denote the forward CIR lengths for $y(t) \to x$ at each time t going forward from $y(t)$.

We agree with the reviewer that these might be somewhat difficult to discern or decode on some screens or in printed format. As such, we have decided to add an additional panel, Panel (c), to Supplementary Figure 4.2 in the SI (Panel (c) from before, "ACI Metric for $y(t) \to x$", is now Panel (d); see also point (4) at the end). This newly added plot serves two purposes:

1. Clearly showcases the objective CIR as a function of t for the time window corresponding to Figure 3.1 in the main text. This is exactly what the reviewer suggested.
2. Additionally showcases the skill of the computationally efficient approximation we have adopted in this work to effectively estimate the objective CIR lengths without resorting to the much more computationally intensive ε-integrals involved in the definition. Specifically, it compares the approximate objective CIR with the exact objective CIR over time t, where each exact CIR was evaluated using a quadrature of the corresponding subjective CIRs over ε.

We hope that with this addition we have not just addressed the reviewer's concerns, but also illuminated further the numerics behind the objective CIR, specifically the efficacious recovery of the objective temporal measure of causal impact through its computationally efficient approximation.

Regards,

Dan Mønster

Reviewer #4

Remarks to the Authors:

Review: Assimilative Causal Inference, by Andreou, Chen and Bollt

► **Authors' Response:** We sincerely thank the reviewer for taking the time to provide a thorough and constructive review of our paper. In what follows, we carefully outline how each of the concerns raised and suggestions given by the reviewer have been addressed and adopted in the revised manuscript. We believe these changes have substantially improved the clarity, rigor, and overall quality of the paper, and for that we again thank the reviewer.

The authors introduce a framework for Assimilative Causal Inference (ACI), which harnesses Bayesian data assimilation to trace the candidate causes of observations. It is demonstrated to be effective for high-dimensional dynamical systems and short datasets, providing real-time tracking of causal roles and establishing a mathematically rigorous criterion for the causal influence range.

The authors describe the general significance of ACI across various scientific disciplines and some of the limitations of existing methods in capturing instantaneous, time-evolving causal relationships in complex systems. The approach is both powerful in scope of application and rigorous in a self-consistent sense and thus should be of interest to a broad scope of researchers interested in complex dynamical systems. In my opinion the paper should be published following a more sober assessment of its limitations and comparison/contrast to other approaches, such as those described in the second paragraph of the introduction.

► **Authors' Response:** We thank the reviewer for their thoughtful assessment of our work. We greatly appreciate their positive feedback regarding the effectiveness, self-consistency, and rigor of our approach, as well as its potential to be of interest to a broad scope of researchers in complex dynamical systems.

For example, I find it intellectually and emotionally distasteful for an author to claim their approach is "paradigm-shifting", as is done in the second sentence of the abstract, upon its introduction? Paradigm shifts require a broad adoption and consensus in the community and years of testing and refinement. This concept was introduced by Thomas Kuhn in his "The Structure of Scientific Revolutions", Univ Chicago Press, 1962. Paradigm shifts in science are not defined by fiat by an author and they take time to evolve. How do the authors know that next week, month or year, etc. another method will not appear that is far superior and supersede all others? Indeed, let the community and time decide what is paradigm shifting.

► **Authors' Response:** We appreciate the reviewer's thoughtful comment regarding our use of the phrase "paradigm-shifting" in the abstract. We fully agree that such terminology should be reserved for developments that have undergone broad community adoption and long-term validation, consistent with Kuhn's original notion. We sincerely apologize for the misuse of this phrase, partially due to the fact that the first two authors are not English native speakers. Our use of the phrase was indeed inappropriate, and we have removed it from the revised manuscript. Thank you again for pointing this out, which helps avoid many potential issues.

The paper highlights the potential of ACI to open new avenues for studying complex systems with transient causal structures and suggests further areas for research and application of ACI. It is one important method of identifying and tracing back causes from observed effects through Bayesian data assimilation and can identify interactions without direct candidate cause observations. It can provide real-time tracking and mathematically validated causal influence range and was shown to be effective in systems showcasing intermittency and extreme events.

► **Authors' Response:** We again thank the reviewer for their positive comments and encouraging assessment of ACI's potential as a new method, not just for temporal causal inference, but also for probing into complex systems through its formulation of the causal influence range. We also appreciate their recognition of how ACI can conduct causal inference naturally, without requiring direct observations from the candidate causes.

As is noted, there are both general limitations and application-specific limitations, that make the paradigm-shifting claims specious. These include:

(1) The fact that the ACI framework may be affected by the correlation between variables during Bayesian data assimilation, which can potentially introduce a spurious causal relationship, even if conditioned over intermediary variables. This can create erroneous interpretations of data.

► **Authors' Response:** Thank you for raising this important question. Let us further clarify the mechanisms behind the conditional ACI framework developed in this work, as it provides a systematic method to resolve spurious correlations and associations when assessing conditional causal relationships. See Section 2.3 of the main text and Section 1.4 of the SI. Conditional ACI differs fundamentally from approaches that condition or marginalize over the (lagged) history of intermediary or ancillary variables that may act as confounders, mediators, moderators, and more. In this sense, it stands in contrast to traditional causal inference approaches such as transfer entropy and causation entropy, as discussed in Section 2.3 and further elaborated in Sections 1.4.3–1.4.4 of the SI.

Unlike many state-of-the-art methods that rely on empirical procedures to remove spurious associations, conditional ACI addresses this issue naturally and self-consistently through its inverse-problem formulation of causality using Bayesian data assimilation. The core ACI structure combines model forecasts with observational information from potential subsequent effects to identify their candidate causes. In conditional ACI, the non-target variables $\mathbf{x}_\mathrm{B}$ continue to influence the underlying system dynamics through the filtering and smoothing equations. However, their observational influence on the estimation of the unobserved candidate causes is explicitly removed. This allows the method to trace the causality of the observed effects back to their genuine causes. Importantly, this procedure preserves the structure and integrity of the governing dynamical equations, distinguishing conditional ACI from many other model-based causal inference techniques.

Regarding the correlations mentioned by the reviewer during the state estimation of $\mathbf{y}(t)$, specifically (a) between $\mathbf{x}_\mathrm{B}$ and $\mathbf{y}(t)$, and (b) between $\mathbf{x}_\mathrm{B}$ and $\mathbf{x}_\mathrm{A}$, the conditional ACI formulation is deliberately constructed to remove these effects. By letting the marginal likelihood uncertainty of $\mathbf{x}_\mathrm{B}$ tend to infinity during the analysis step, formally $\mathrm{Var}(\mathbf{x}_\mathrm{B}(t)) \to +\infty$, the corresponding correlations are effectively ignored or resolved when inferring the conditional causal relationship $(\mathbf{y}(t) \to \mathbf{x}_\mathrm{A}) \,|\, \mathbf{x}_\mathrm{B}$. The resolution of (a) through this mechanism has been discussed in Section 2.3 of the main text and Section 1.4 of the SI. Thanks to the reviewer's suggestion, we have added more discussions in Section 2.3 of the main text to clarify this argument. We provide the relevant excerpt here:

"In the more general context of a complex system, state variables can be grouped into three disjoint subsets: $\mathbf{x}_\mathrm{A}$, $\mathbf{x}_\mathrm{B}$, and $\mathbf{y}$, with $\mathbf{x} = (\mathbf{x}_\mathrm{A}, \mathbf{x}_\mathrm{B})^\mathtt{T}$ under the previous notation. The goal is to determine whether $\mathbf{y}$ is a cause of $\mathbf{x}_\mathrm{A}$ in the presence of the remaining variables $\mathbf{x}_\mathrm{B}$, which are called the non-target variables, including the cases of confounders or mediators. Since $\mathbf{x}_\mathrm{B}$ interacts with both $\mathbf{x}_\mathrm{A}$ and $\mathbf{y}$, appropriately accounting for the role of $\mathbf{x}_\mathrm{B}$ while inferring the causal link from $\mathbf{y}$ to $\mathbf{x}_\mathrm{A}$ becomes essential. Hereafter, let us assume that a realization for each of $\mathbf{x}_\mathrm{A}$ and $\mathbf{x}_\mathrm{B}$, namely $\mathbf{x}_\mathrm{A}(s \leq T)$ and $\mathbf{x}_\mathrm{B}(s \leq T)$, is available. Therefore, the candidate cause $\mathbf{y}$ is the only variable that does not require to have observations in this framework. Following the argument in Section 2.1, the causal relationship

from $\mathbf{y}(t)$ to $(\mathbf{x}_\mathrm{A}, \mathbf{x}_\mathrm{B})$ is identified based on the uncertainty reduction in the smoother distribution p_t^s related to the filter one p_t^f, where

$$p_t^\mathrm{f} = p(\mathbf{y}(t)|\mathbf{x}_\mathrm{A}(s \le t), \mathbf{x}_\mathrm{B}(s \le t)),$$
$$p_t^\mathrm{s} = p(\mathbf{y}(t)|\mathbf{x}_\mathrm{A}(s \le T), \mathbf{x}_\mathrm{B}(s \le T)).$$

The main focus here is to appropriately determine and resolve the presence of $\mathbf{x}_\mathrm{B}$ from p_t^f and p_t^s as to infer a conditional causal relationship from $\mathbf{y}(t)$ to $\mathbf{x}_\mathrm{A}$ beyond the contributions of $\mathbf{x}_\mathrm{B}$.

*Recall from Section 2.1 that ACI employs Bayesian data assimilation to estimate the state of $\mathbf{y}(t)$. ACI exploits observations of $\mathbf{x}$ to reduce the uncertainty in $\mathbf{y}(t)$, where the influence of $\mathbf{x}$ on the estimation of $\mathbf{y}(t)$ depends on its own uncertainty: smaller observational uncertainty leads to a stronger impact [**45**]. When extending this framework to $\mathbf{x} = (\mathbf{x}_\mathrm{A}, \mathbf{x}_\mathrm{B})^\mathtt{T}$ for conditional ACI, the observed time series of $\mathbf{x}_\mathrm{B}$ is still used in the forecast step to compute the uncertainties in $\mathbf{x}_\mathrm{A}$ and $\mathbf{y}$. However, during the analysis step, the uncertainty in $\mathbf{x}_\mathrm{B}$'s dynamics is deliberately excluded by assigning infinite uncertainty to its marginal likelihood before posterior computation. This can be rigorously formulated through a limit operation on the analysis component of the Bayesian data assimilation update equations, which formally leads to the following filter- and smoother-based distributions:*

$$p_t^{\mathrm{f}|\mathbf{x}_\mathrm{B}} := \lim_{\mathrm{Var}(\mathbf{x}_\mathrm{B}(t)) \to +\infty} p_t^\mathrm{f}, \quad p_t^{\mathrm{s}|\mathbf{x}_\mathrm{B}} := \lim_{\mathrm{Var}(\mathbf{x}_\mathrm{B}(t)) \to +\infty} p_t^\mathrm{s}.$$

Similar to (2.5), if the relative entropy of $p_t^{\mathrm{s}|\mathbf{x}_\mathrm{B}}$ from $p_t^{\mathrm{f}|\mathbf{x}_\mathrm{B}}$ is positive, $\mathcal{P}\big(p_t^{\mathrm{s}|\mathbf{x}_\mathrm{B}}, p_t^{\mathrm{f}|\mathbf{x}_\mathrm{B}}\big) > 0$, then $\mathbf{y}$ is identified as the cause of $\mathbf{x}_\mathrm{A}$ conditional to $\mathbf{x}_\mathrm{B}$ at time t. This means future data of $\mathbf{x}_\mathrm{A}$ incur additional information gain about $\mathbf{y}(t)$ related to its history, beyond the observational contributions of $\mathbf{x}_\mathrm{B}$. By computing this metric at each t, a time-dependent conditional causal relationship is established."

For (b), the details are somewhat more subtle, and we acknowledge that the SI may not have fully elaborated on this point. While we alluded to the handling of correlations in the observed state space during the proof of the principle of nil conditional assimilative causality for CGNSs in Section 3.2 of the SI, we did not restate this explicitly in Section 1.4. In essence, by sending $\mathrm{Var}(\mathbf{x}_\mathrm{B}(t)) \to +\infty$ during analysis, the correlations between $\mathbf{x}_\mathrm{A}$ and $\mathbf{x}_\mathrm{B}$ are also driven to zero at that time t. Consequently, observational correlations are simultaneously resolved during the state estimation of $\mathbf{y}(t)$, ensuring that within the ACI framework the observed subsequent effects $\mathbf{x}_\mathrm{A}$ are traced back to their genuine candidate causes $\mathbf{y}(t)$. In this regard, conditional ACI can be loosely compared to assigning an improper or diffuse prior in parametric Bayesian inference. By assigning infinite uncertainty to the marginal likelihood of $\mathbf{x}_\mathrm{B}$ during the analysis step, $\mathbf{x}_\mathrm{B}$ effectively "uncorrelates" from $\mathbf{x}_\mathrm{A}$, thereby allowing only the future information of $\mathbf{x}_\mathrm{A}$ to inform the state estimation of $\mathbf{y}(t)$ beyond its own history. Following the reviewer's concern, we briefly summarize the discussion regarding (b) from above and include it in the following revision of point **(1)** from Section 1.4.3 in the SI.

"Complete elimination of $\mathbf{x}_\mathrm{B}$'s influence during data assimilation, ensuring any uncertainty reduction in $\mathbf{y}(t)$ stems solely from $\mathbf{x}_\mathrm{A}$, consistent with our ACI objectives. Furthermore, during analysis, the correlations between $\mathbf{x}_\mathrm{A}$ and $\mathbf{x}_\mathrm{B}$ in their likelihood are necessarily sent to zero as the uncertainty of $\mathbf{x}_\mathrm{B}$ goes to infinity. This further resolves the potential for spurious associations arising over the observable state space that can flow over when inferring causality from $\mathbf{y}(t)$ to $\mathbf{x}_\mathrm{A}$; see Theorem 3.2. A rough analogue to this approach would be the assignment of an improper prior as an uninformative or diffuse prior in parametric Bayesian inference."

(2) The real-time tracking capability of the ACI framework necessitates continuous observation of a dataset. This requirement can be challenging in systems where data is not continuously available or is incomplete, which is often the case.

▶ **Authors' Response:** Thank you for bringing this important question up. Actually, we wrote in Section 2.1.1 of the main text, *"For simplicity, the time series are assumed to be continuously observed. An analogous framework can be developed for discrete-in-time observations"*, and in Section 1.1 of the

SI, *"For convenience, the time series of* $\mathbf{x}$ *is assumed to be continuously observed, with the development of an analogous framework for discrete-in-time observations being possible".* The procedure of the ACI framework can be naturally adopted to the cases where the observations of $\mathbf{x}$ are obtained at discrete time intervals. The same applies to the assumption of non-uniform or non-constant observational rates, which can lead to incomplete datasets. We acknowledge that this point may not have been sufficiently emphasized in the original version. In response to the reviewer's suggestion, we have added the following paragraph in the discussion section of the main text to highlight this aspect:

"Finally, continuous-in-time observations are adopted in this work for simplicity of presentation. Nevertheless, an analogous framework can be naturally extended to discrete-in-time observations using the corresponding Bayesian data assimilation algorithms. An interesting scientific question in this context is to understand the uncertainty in causal detection arising from the discrete nature of the data, particularly at time intervals between successive observations. This line of inquiry could also highlight an important distinction between ACI and purely data-driven methods. In ACI, the dynamical model provides a natural way to interpolate causal relationships at time instants when observations are unavailable. Therefore, it may help bridge observational gaps and maintain temporal continuity in causal inference."

We have also implemented a new test using real-world observations that are noisier and which also introduces model and discretization errors into the framework. This is in Section 4.3.3 of the SI. We have added the following at the end of the results section:

"In addition, in the Supplementary Information, a case study using real-world ENSO observations is examined, which does not assume a perfect model setup. The conclusions derived about the causal relationships using real-world observational data remain qualitatively similar to those with model-generated data. The results provide preliminary evidence that the ACI framework remains reasonably robust under realistic conditions, including observational noise, moderate model error, and numerical discretization effects."

One supporting evidence toward a discrete-in-time formulation in this work is illustrated by the use of the discrete-in-time online smoother for CGNSs to analytically compute the CIRs in the case studies (see Sections 2.2–2.3 of the SI). This algorithm can accommodate $\mathbf{x}$ observed at discrete, possibly non-uniform, intervals. Nevertheless, the use of discrete observations for $\mathbf{x}$ can also be naturally implemented in ACI through consistent discretizations of the posterior filter and smoother state estimation equations. For the general stochastic complex systems assumed in this work (see (2.1) in the revised main text and (1.1) in the SI), these ideas naturally extend to that case too. Specifically, we can leverage the powerful results of Bayesian data assimilation concerning the filter and smoother state estimation equations that track the evolution of the associated PDFs through exact random PDEs which we provide in their unnormalized form in (1.8) and (1.11) in the SI, respectively. In the case where $\mathbf{x}$ is observed at discrete time intervals, these equations (or any of their approximations, e.g., ensemble variations) can be discretized in time through an appropriate numerical model, with the continuous-in-time ACI formulation giving rise to a discete one completely analogously. The same result also applies to the CIR analysis through appropriately discretized online smoother formulations, as already described in Section 2.3 for CGNSs.

(3) Although ACI claims to scale efficiently to high dimensions, the complexity of the framework still poses significant computational challenges, particularly in very large-scale systems.

► **Authors' Response:** Thank you for raising this important question. As discussed in the introduction, since ACI is built upon Bayesian data assimilation, it naturally benefits from the many numerical methods already developed in that field to handle high-dimensional systems. Therefore, ACI can, in principle, be scaled to address such problems as well. The primary goal of this paper is to develop the ACI framework and to distinguish it from existing methods, both in terms of its methodology and the new features it reveals. For this reason, we have focused on relatively lower-dimensional systems for numerical illustrations. Extending the framework to much higher-dimensional systems is conceptually straightforward, although careful numerical design is required, which is an effort that could merit one or more separate studies for full implementation. Nevertheless, we appreciate the reviewer's insightful comment and have added a paragraph in the discussion section outlining a more concrete path toward applying ACI to high-dimensional systems.

*"Second, by building on the computationally efficient methods developed in Bayesian data assimilation, it is essential to apply ACI to higher-dimensional systems. Ensemble data assimilation is a well-established technique that routinely uses localization, covariance inflation, and other adjustments to overcome the curse of dimensionality [**54, 55, 56**]. As a result, the ensemble size (typically on the order of $O(100)$) is often chosen independently of the system's dimension, which contributes to the scalability of Bayesian data assimilation and, consequently, of ACI. However, the sampling errors inherent in ensemble methods may introduce additional biases. Thus, it is necessary to examine whether such errors affect ACI. In addition, various coarse-grained approaches, such as approximating the full distribution with a Gaussian, are widely used in practice. Although these approximations introduce errors in computing the PDF, they are robust and highly accurate when computing diagnostic indicators, such as the relative entropy, that are used for model identification and sensitivity analysis. Thus, they may substantially accelerate ACI in practice."*

(4) The framework assumes the underlying models are correctly specified without observational noise, a condition that is rarely true in real-world applications where measurements are often noisy in many distinct, but difficult to quantify, ways.

► **Authors' Response:** Thank you for bringing up these valid concerns. These are indeed crucial practical issues for all causal inference methods. While detailed investigations would warrant one or more dedicated papers, we appreciate the reviewer's insightful comments and have expanded the discussion along these lines in the revised manuscript. First, the Bayesian data assimilation framework underlying ACI does not require a perfect model. Thus, a misspecified model can be naturally incorporated within ACI to study its effects. By performing two parallel experiments (one using a perfect model and another using an imperfect model), we can directly assess the impact of model error on causal inference. More importantly, based on the reviewer's insights, ACI itself can be leveraged to detect model error and to help construct parsimonious, data-informed dynamical models. Although this extends beyond the core goal of causal inference, it represents a broader contribution to model identification and the dynamical systems community. Second, because ACI is formulated within the Bayesian data assimilation framework, it can also address the influence of observational noise. A natural strategy is to apply a pre-filtering step to mitigate part of the observational noise before conducting ACI. An even more informative study is to compare the ACI results obtained from the original noisy data with those from the pre-filtered data. The differences between the two provide a valuable diagnostic of how observational noise affects causal inference. Since Bayesian data assimilation is already integral to ACI, such pre-filtering can be performed using model-based information rather than empirical noise-removal methods, as required in many other approaches. Overall, these directions represent important and promising topics that merit careful and systematic investigation in future studies. We have added the following paragraph in the discussion section to summarize the ideas of using ACI to study these crucial topics:

"First, because model error is ubiquitous in practice, it is essential to study its impact on causal inference. Understanding how model error affects inferred causal relationships can create opportunities not only to detect and correct model error, but also to leverage this information to develop parsimonious models for complex dynamical systems. Similarly, observational data may contain additional noise that can affect causal inference. The ACI framework provides a natural way to analyze how observational noise contaminates causal detection. In such a context, data assimilation can serve as a pre-filtering step to reduce the influence of observational noise, which allows for a systematic examination of its role in shaping causal inference outcomes."

(5) While the authors noted the applicability in short datasets, the accuracy and reliability of causal inferences drawn from very limited data remain a concern.

► **Authors' Response:** We thank the reviewer for raising this crucial question. Short datasets indeed pose significant challenges for most data-driven causal inference methods. However, because ACI integrates model information with observational data, it requires only a single data point at each time instant to identify instantaneous causal relationships. This represents one of the fundamental distinctions between ACI and standard data-driven approaches. We thank reviewer again for this comment, which helps us improve our presentation. To this end, we have clarified this difference in the discussion section:

"Finally, ACI requires only a single time series but assumes access to a dynamical model, which enables it to identify instantaneous causal relationships. In contrast, most time-series-based methods operate under model-free assumptions and therefore focus on aggregated or time-averaged causal links rather than on causality at specific moments. Because ACI incorporates model dynamics, it does not require long time series to detect causes and effects. Instead, it can infer instantaneous causal links from a single data point combined with model information. In contrast, most purely data-driven methods rely on long time series and can only reveal causal relationships in a time-averaged sense."

(6) The application-specific limitations include (a) extreme event analysis, where accurate parameter selection and model calibration are essential; (b) historical data limitations, such as shifts not represented in a dataset and (c) interference handling, which can compromise the reliability of the inferred causal relationships.

► **Authors' Response:** We thank the reviewer for raising this thoughtful question. If our understanding is correct, the reviewer is referring to the accuracy of the model used in the extreme event study and, in general, the ACI framework. Indeed, this accuracy is essential. In our analysis of extreme events, we employed a perfect-model setup, which we have clarified in the revised version of the main text. The term perfect model does not imply that the model reproduces the exact trajectory of the observations. Because the underlying system is chaotic and subject to stochastic forcing, a perfect model means that the model used in ACI shares the same governing equations and long-term statistical behavior as the system that generates the true signal. Accurate parameter estimation and model calibration are therefore crucial. The historical data used for training should not be statistically biased, although it is not necessary for the exact events in the testing period to appear in the training dataset. Together with the response to question (4) of the reviewer, we have expanded the discussion section to include an explicit consideration of model error and its implications for ACI:

"First, because model error is ubiquitous in practice, it is essential to study its impact on causal inference. Understanding how model error affects inferred causal relationships can create opportunities not only to detect and correct model error, but also to leverage this information to develop parsimonious models for complex dynamical systems."

While these limitations are noted, and notable, the reader will strongly benefit from knowing how these few, of potentially many more, are treated in other approaches in the field, many of which are mentioned in the introduction and are known to the authors in more detail.

► **Authors' Response:** We again thank the reviewer for taking the time to provide this constructive review with crucial feedback, and valid and important concerns. We have added discussions and clarifications to explain the issues raised by the reviewer. With the implementations and changes made here based on the reviewer's suggestions, we wish we have improved the quality of the manuscript.

We sincerely thank all of the reviewers for their time and suggestions!

Additional changes independent of the Reviewers' comments:

(1) In the figures corresponding to the ENSO case study (Figure 3.2 in the main text and Supplementary Figures 4.5–4.8 in the SI), each of the ENSO event color legends on the y-axis (or x-axis accordingly) has been changed to now range from September to February, which coincides with the peak of the actual event during boreal winter. Furthermore, we have slightly increased the threshold for defining extreme EP El Niño events (increased by 0.5°C for the average during boreal winter December–January–February), as to capture actual significant extreme events and not marginally stronger moderate EP El Niño events. Of course, these changes do not alter any of the numerics present in the initial submission, since they are purely cosmetic in nature. The goal of these changes is to improve figure readability and interpretability.

(2) Minor changes have been made in Panels (f)–(g) of Figure 3.2 in the main text and the Supplementary Figures 4.5–4.6 in the SI, which lead to only a slight change to the temporal profile of the ACI measure, which do not alter or change any of the conclusions made in this case study.

(3) Minor illustration changes have been made to the figures in the SI. Specifically we have increased some font sizes and centered titles & labels in these plots for readability.

(4) In Supplementary Figure 4.2, beyond the additions made in line with Reviewer #3's suggestions (added Panel (c), which illustrates the temporal profile of the definition and computationally efficient approximation for the objective CIR), we have also significantly increased the ε resolution of the subjective CIR plot (Panel (b) in the revised figure) for clarity and for better illustration.

(5) Slightly adjusted some cosmetic details and added clarifications in the schematic diagram figure (Figure 2.1 in the main text), specifically Panels (a) and (c), to help readers obtain a more intuitive idea of both ACI and its conditional generalization.

(6) We have corrected minor typos and rephrased certain sentences for clarity and readability throughout the manuscript (main text and SI). All of these changes are clearly noted in the attached `diff` files.

(7) We have also added brief additional clarifications and comments to the parts that needed it for exposition's sake (main text and SI). All of these additions are clearly noted in the attached `diff` files.

(8) Finally, we have also made very minor notational changes to some equations for readability (e.g., adjusted parentheses sizes and replaced incorrect use of τ with $\boldsymbol{\tau}$ for the subjective and objective CIR lengths at some instances in the main text and SI). All of these changes are clearly noted in the attached `diff` files and are purely cosmetic in nature.

Response to Reviewers

Ref.: Response to Referees
Manuscript Number: NCOMMS-25-48940A
Manuscript Title: Assimilative Causal Inference
Authors: Marios Andreou; Nan Chen; Erik Bollt
Corresponding Author: Nan Chen

Dear Reviewers:

We deeply appreciate the reviewers for taking the time to thoroughly review our work and provide such thoughtful and insightful feedback. The reviewers' detailed and discerning eye helped us clarify several valuable points and implement additions and revisions that will further aid readers in understanding and appreciating our framework. We sincerely thank the reviewers for their time, rigor, insights, and endorsement through the peer review process.

The revised manuscript and supplementary information are attached for your consideration. In the following pages, we provide a detailed, point-by-point response to the reviewers' comments, outlining how each has been thoughtfully considered in the revised manuscript.

Point-by-Point Response to Reviewers' Comments:

Reviewer #1

Remarks to the Authors:

I would like to thank the authors for their detailed response and changes. I believe that the new version of the paper is substantially better.

► **Authors' Response:** We thank the reviewer very much for their positive feedback and for all their constructive suggestions, which have significantly improved the clarity, rigor, and quality of our manuscript. We sincerely thank them for taking the time to review our manuscript.

However, some of my concerns remain:

Regarding claims of scalability: I appreciate the authors' clarification that ACI is designed to leverage existing Bayesian data assimilation machinery and hence their method can be applied in high-dimensional settings. However, such clarification is not well supported. The rebuttal does not provide a concrete complexity analysis of the composite ACI method in terms of state dimension, nor does it include new experiments on higher-dimensional systems. The current experiments still only contain low-dimensional examples (up to six variables), which does not convincingly substantiate the strong claim that the method "scales efficiently to high-dimensional problems."

I would therefore encourage the authors, at minimum, to soften the wording of the scalability claim throughout the paper. Perhaps the authors can emphasize that ACI can in principle be implemented in high-dimensional settings by reusing efficient DA algorithms, while leaving a full empirical demonstration and more detailed complexity analysis to future work.

► **Authors' Response:** Thank you very much for the detailed comments. We have followed the reviewer's suggestion to soften the wording regarding the scalability of the method to high dimensions. Specifically, we modified the abstract by adopting what the reviewer suggested:

"It uniquely identifies dynamic causal interactions without requiring observations of candidate causes, accommodates short datasets, and, in principle, can be implemented in high-dimensional settings by employing efficient data assimilation algorithms."

We have also adjusted the relevant sentence in the Discussion section following the same principle. Later in the conclusion, we include a paragraph discussing a more detailed implementation of how Bayesian data assimilation can help scale ACI to high-dimensional problems. This has been assigned as part of future work. Thus, the tone concerning the method's scalability has been softened over the entire manuscript.

"By interpolating causality from effects to causes instead of extrapolating the latter forward-in-time, ACI offers an inherently more robust framework compared to classical predictive causality."

While I understand the key difference in approach, I do not see how it is apparent that "interpolating causality from effects to causes" results in an "inherently more robust framework". Without sufficient theoretical or empirical justification, especially reading up to the point of the paper where the claim is made, it is hard to validate it.

► **Authors' Response:** Thank you very much for pointing this out. We did not express ourselves clearly on this point. In the revised main text, we have added the following sentence after the quoted excerpt as to provide further explanation. We have used the word "typically", hoping to provide general empirical evidence rather than a rigorous claim:

"Interpolation is typically more stable because it operates within the range of observed information, whereas extrapolation requires projecting beyond available data and is therefore far more sensitive to model error and uncertainty."

Regarding real-world experiments: I appreciate the authors' response highlighting the difficulty of working with issues like addressing model error and sampling biases. I also believe that the new experiments with the real-world ENSO data have made the paper stronger.

However, I would still encourage the authors to add a discussion to compare the experimental results of their method to the baselines on the real-world ENSO dataset to highlight scenarios where their method would perform more favorably compared to the baselines. While the theoretical contributions of the paper are indeed quite significant, practically demonstrating the ability of ACI to discover instantaneous edges and the resulting improvements for causal discovery performance, would further strengthen the contribution of the paper.

► **Authors' Response:** Thank you very much for the suggestion. We have added the following new paragraph to Section 4.3.3 of the supplementary information ("ACI on Real-World ENSO Data"):

*"Finally, it is worth emphasizing the distinction between ACI and many baseline causal inference methods for analyzing real-world ENSO datasets [**27, 28, 29**]. Most of the current approaches are purely data-driven and focus on identifying broad, time-averaged cause-and-effect relationships. Their findings are generally consistent with physical mechanisms, such as the causal influence of the thermocline depth on the SST and the discharge–recharge loop linking El Niño and La Niña. Yet, ACI provides a more refined and event-specific indicator that reveals how individual variables contribute to the causal structure of each ENSO episode, an aspect often missing from state-of-the-art techniques. ACI yields markedly different causal strengths and CIR patterns across events, providing valuable insight into instantaneous, time-localized cause-and-effect behavior. As a natural direction for future work, the ACI framework can be coupled with a range of ENSO models and expanded to include additional variables, thereby enhancing causal discovery with a particular focus on instantaneous relationships."*

Reviewer #2

Remarks to the Authors:

I co-reviewed this manuscript with one of the reviewers who provided the listed reports. This is part of the Nature Communications initiative to facilitate training in peer review and to provide appropriate recognition for Early Career Researchers who co-review manuscripts.

► **Authors' Response:** We sincerely thank the reviewer again for their efforts and constructive feedback provided under the Nature Communications co-review initiative for early career researchers. Their input and time is greatly appreciated.

Reviewer #3

Remarks to the Authors:

Dear editor and authors,

This is my response to the revised manuscript entitled "Assimilative Causal Inference."

I want to thank the authors for a very thorough, detailed and easy-to-read response letter and for providing diff-files making it clear what has been changed.

After reviewing the revised manuscript, I am satisfied that my initial comments have all been adequately addressed by the authors. Specifically, I am happy to see that the authors have provided a repository with the code used to produce the results in the manuscript.

I recommend publication of this revised manuscript, and I look forward to more research on the method and to see applications by other researchers.

Regards,
Dan Mønster

► **Authors' Response:** We are grateful to the reviewer once again for their insightful and constructive comments, which have greatly enhanced the clarity, rigor, and quality of our manuscript. We sincerely thank them for taking the time to review our manuscript.

Reviewer #4

Remarks to the Authors:

I thank the authors for their thorough response to my report and those of the other referees and the associated revisions of the text, and am pleased to recommend the paper for publication.

► **Authors' Response:** We thank the reviewer once again for their highly constructive feedback and suggestions, which have significantly improved the clarity, rigor, and quality of our manuscript. We sincerely thank them for taking the time to review our manuscript.

We sincerely thank all of the reviewers for their time and suggestions!